CHAPTER 3: IMPLEMENTATION OF THE SECURED IP TELEPHONY SYSTEM.

3.1: STUDY OF YARO CONSULTING TELEPHONE SYSTEM

ETS YARO CONSULTING being a consulting company having a variety of data and information to manage in the petroleum sector, it has a variety of telephone equipment for communication within its premises.

A PBX (Private Branch Exchange)

This Panasonic KX-NS700UEX equipment integrates analog and digital telephony functions. This PBX acts as a telephone server and is the essential central equipment for analog telephony.

• 2 Panasonic digital telephones

• 6 Panasonic analog telephones

3.2 IT Equipment and The Network Architecture of ETS YARO CONSULTIG

3.2.1 ETS YARO CONSULTING IT EQUIPMENT

They include:

• A computer rack consisting of a switch, two patch panels, a DVR for surveillance cameras, and a PBX for telephony.

• Seven computers distributed throughout the company's various offices.

a. Existing IT architecture of YARO COBSULTING

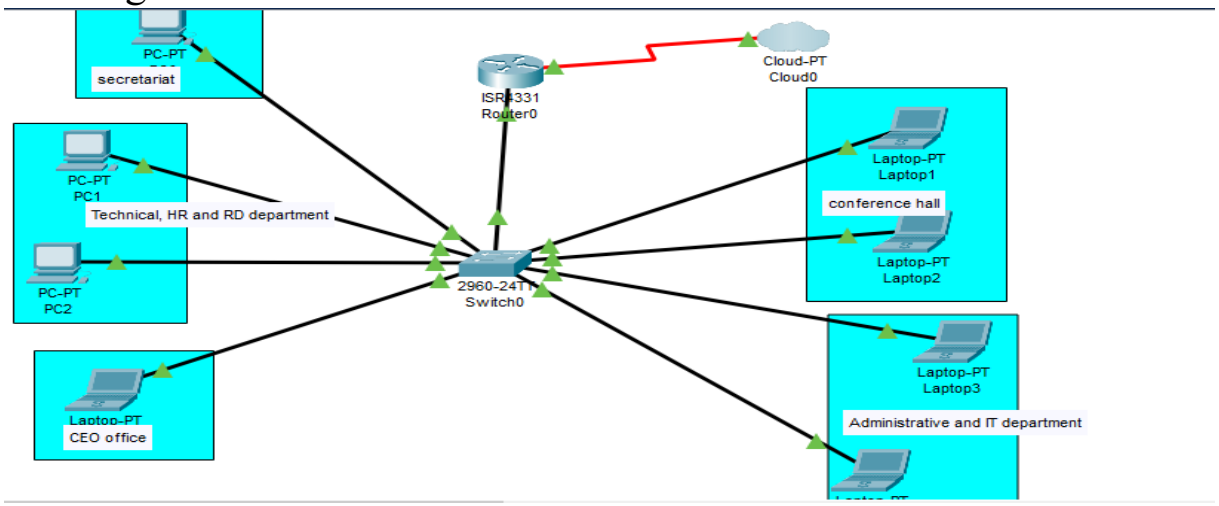


figure 3.2.1 a 1: Architecture of Yaro consulting

b. PROPOSED ARCHITECTURE

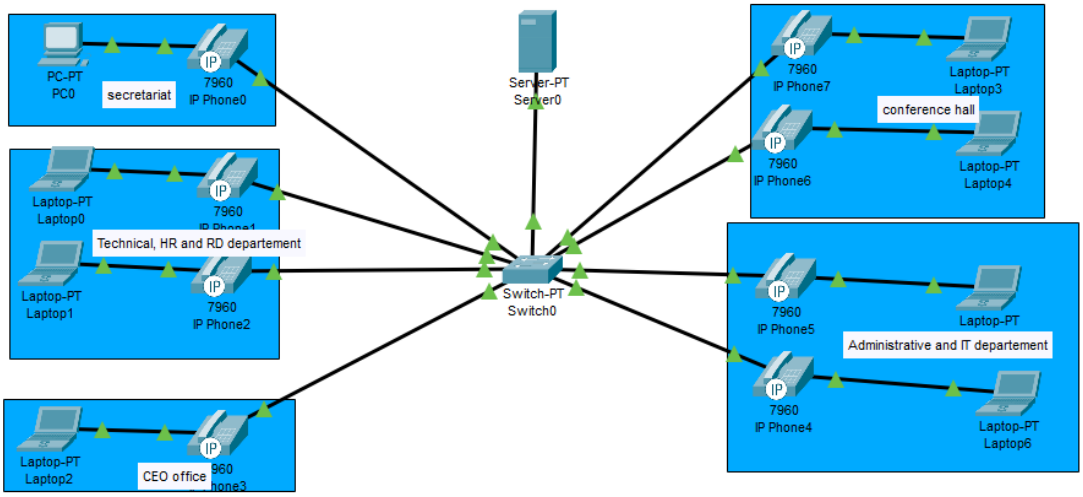


figure 3.2.1 a 2: Proposed Architecture for ETS YARO CONSULTING

With this architecture, we no longer need two cable lines, one for IT and one for telephony. With IP telephony, a single line is sufficient for both services because the IP phone has two RJ45 ports, one dedicated to the switch and the other dedicated to connecting to a computer with an RJ45 port. To implement this architecture, we must purchase an IPBX because the existing PBX cannot support VoIP.

Since IPBX prices vary between €2,000 and €3,000, we will purchase an IPBX for €2,000.

3.3: Installing ELASTIX and Configuring the Phone Service

3.3.1 OVERVIEW OF THE DIFFERENT TOOLS TO USE

1- The ELASTIX Server

ELASTIX is a free and proprietary private branch exchange (PABX) for GNU/Linux systems. It offers, among other things, voicemail, queues, call agents, music on hold and call park, and call distribution. It is also possible to add conferencing through the installation of additional modules and recompiling the binaries.

ELASTIX includes a very large number of features allowing complete integration to meet the majority of telephony needs. It allows you to completely replace a proprietary PABX using FXO/FXS cards, and to add VoIP features to transform it into an IP PBX. It also allows full VoIP operation using commercially available SIP or IAX phones. Finally, call routing, voice menu, and voicemail features, among others, place it at the level of the most complex PBXs. In large ELASTIX installations, it is common to deploy functionality across multiple servers.

One or more central processing units will be dedicated to call processing, with auxiliary servers handling secondary tasks (such as a database, voicemail, and conferencing). Asterisk also offers several other services beyond phone service, including:

- The messaging services
- Security and encryption.

2- The Wi-Fi access point

Here, the access point plays a very important role for wireless users. It is connected to the corporate network and interconnects users with smartphones and computers without network cards.

3- The terminals

The terminals are analog phones, IP phones, and softphones. They act as communication devices.

Note: Analog phones connect to the IP network using Linksys adapters, which adapt the ports of analog phones to the IP network.

3.3.2 INTRODUCTION TO ELASTIX SERVER



figure 3.3.2 1 : Logo of Elastix

There are several VoIP solutions, but we chose to work with the Asterisk server because it's hosted on a Linux server system, which requires some familiarity with Linux systems, because of its more advanced security compared to other VoIP servers, because of its slightly more complex configuration, and because of its easy administration.

Asterisk installation is generally done from the command line in the terminal of a Linux distribution. We work with Ubuntu because it's a system dedicated to server-type configurations, and everything is done in the terminal except when it acts as a potential client so we can access its interface.

Demonstration of installation of ELASTX

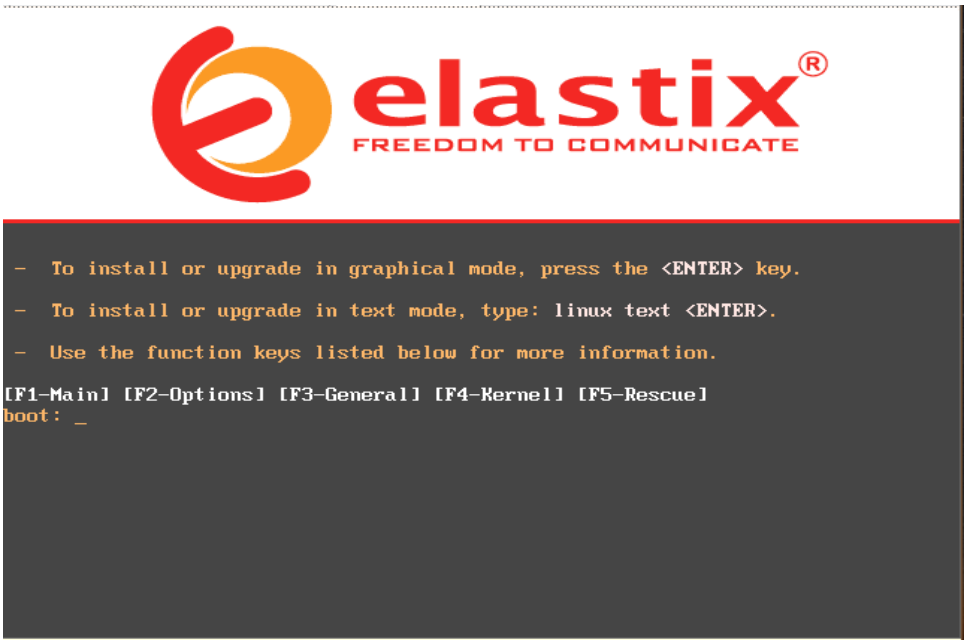


figure 3.3.3 a 1 booring elastix

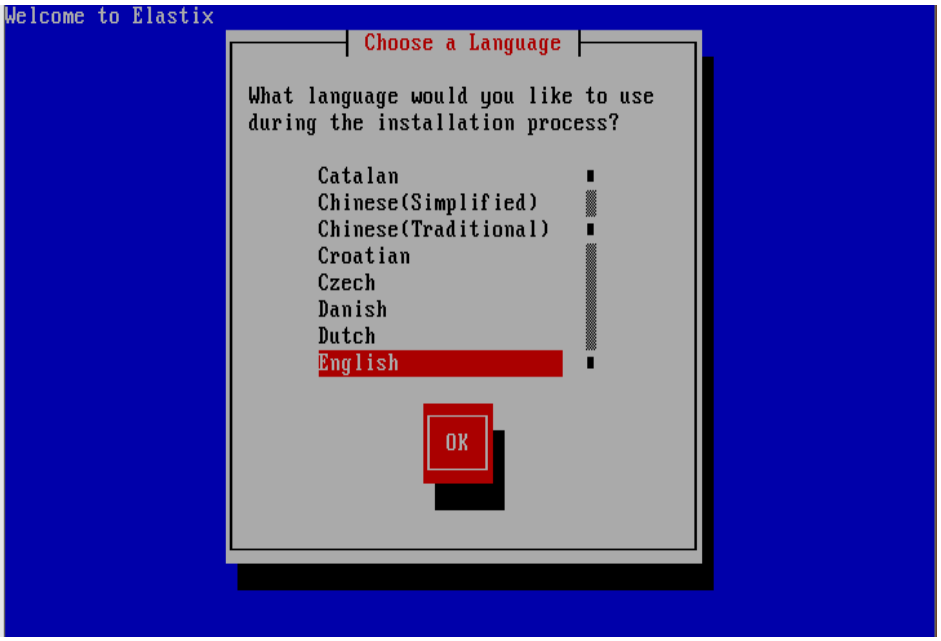


figure 3.3.3 a 2: CHOOSE A LANGUAGE

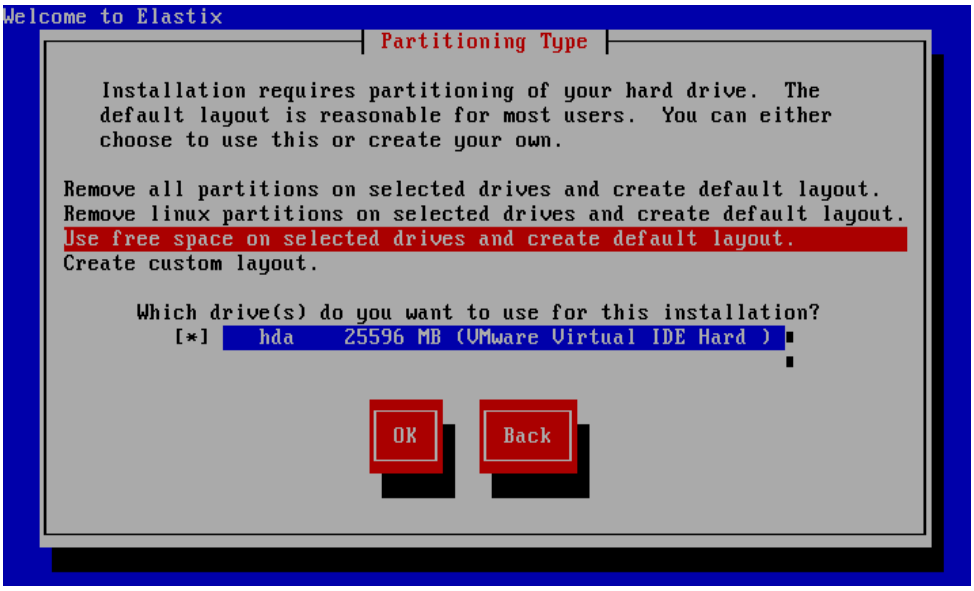


figure 3.3.3 a 3 : CHOOSE DISK SIZE AND PARTITION

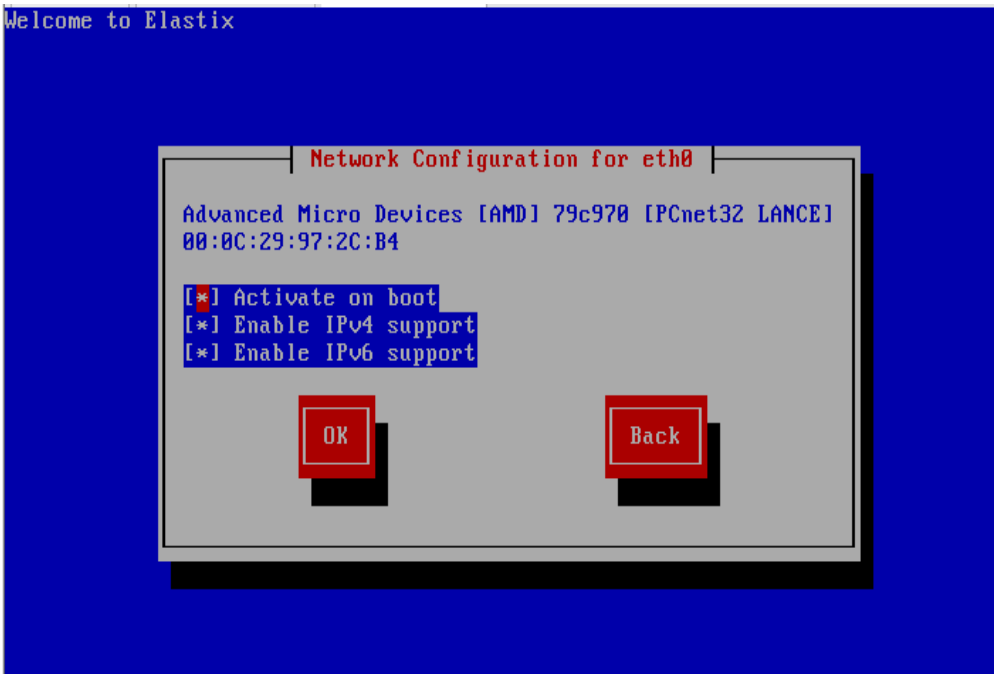


figure 3.3.3 a 4 : NETWORK INTERFACE CONFIGURATION

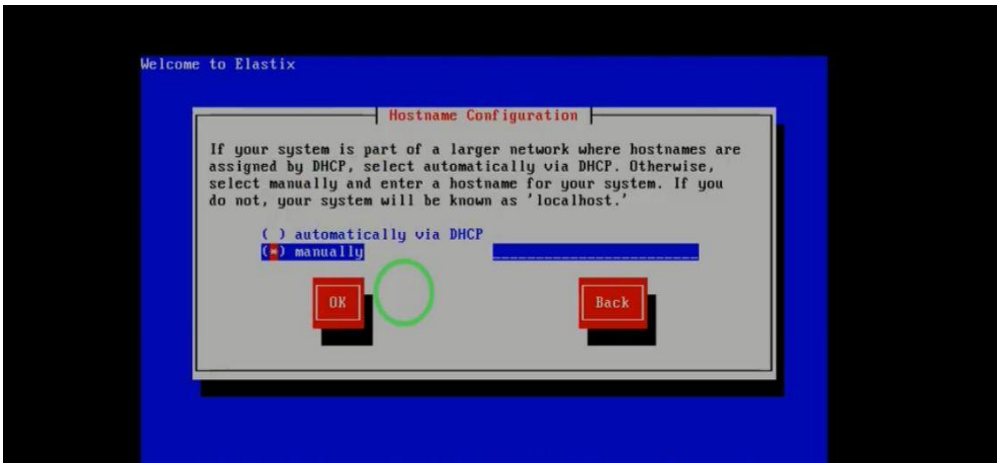


figure 3.3.3 a 5: HOST NAME CONFIGURATION

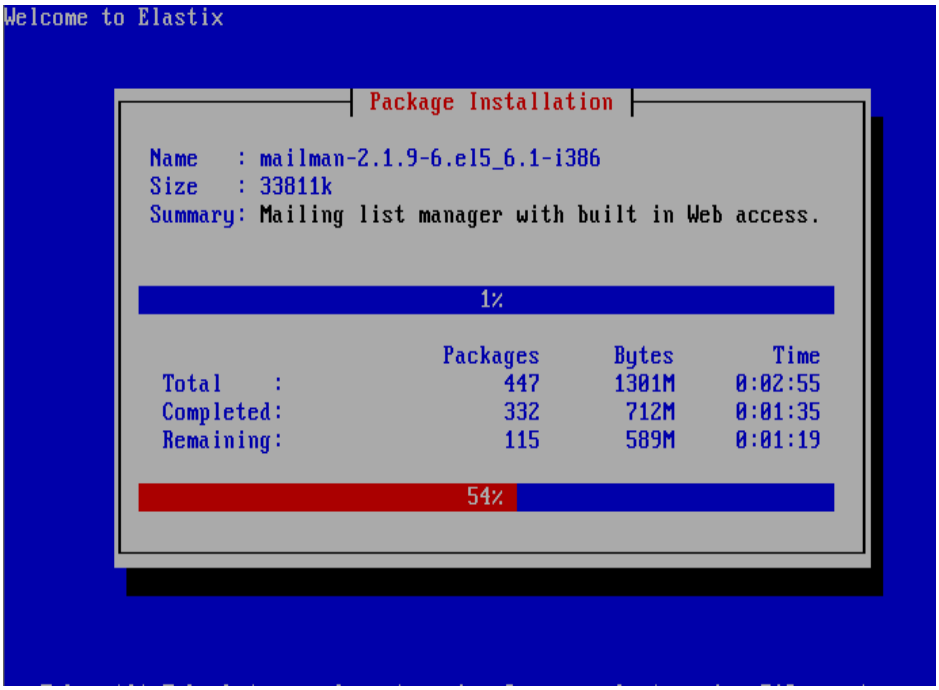


figure 3.3.3 a 6: PACKETS INTALLATION

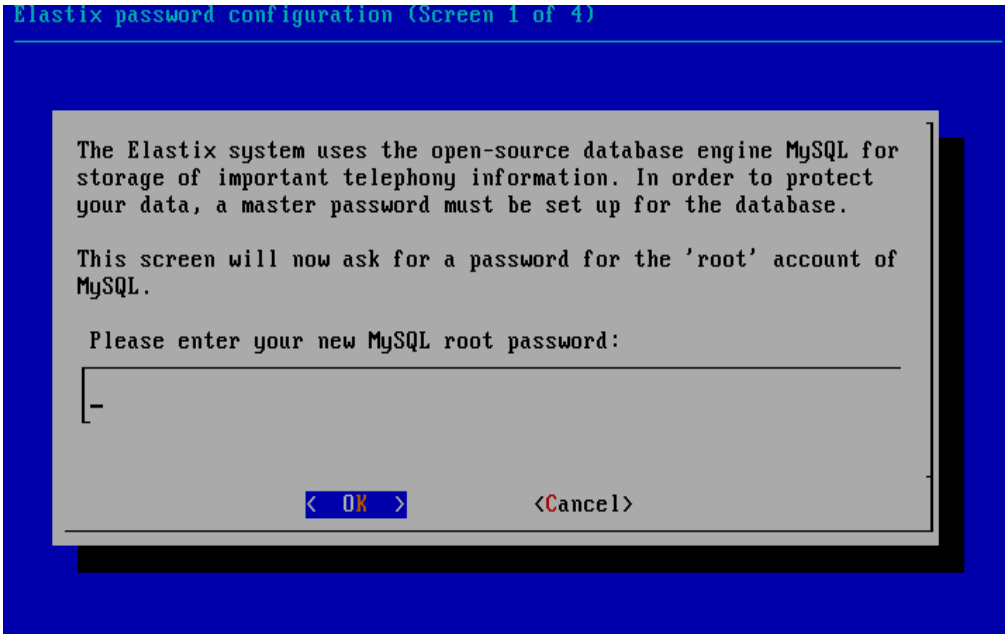


figure 3.3.3 a 7 : DATABASE MYSQL CONFIGURATION

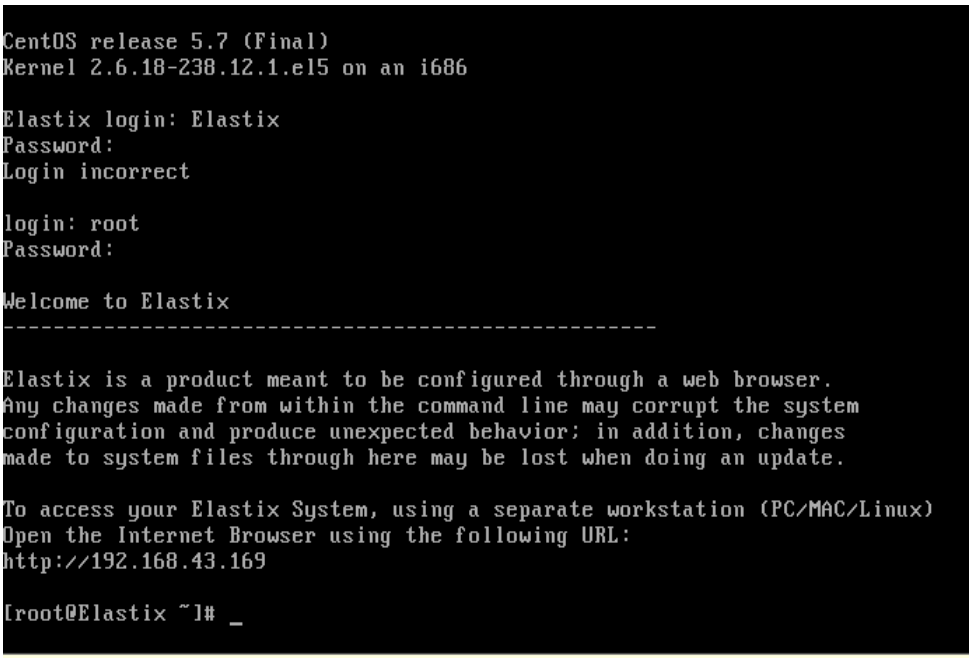


figure 3.3.3 a 8 : CONNECTING TO ELASTIX SERVER

```
Welcome to Elastix
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Elastix is a product meant to be configured through a web browser.
Any changes made from within the command line may corrupt the system
configuration and produce unexpected behavior; in addition, changes
made to system files through here may be lost when doing an update.

To access your Elastix System, using a separate workstation (PC/MAC/Linux)
Open the Internet Browser using the following URL:
http://192.168.43.169

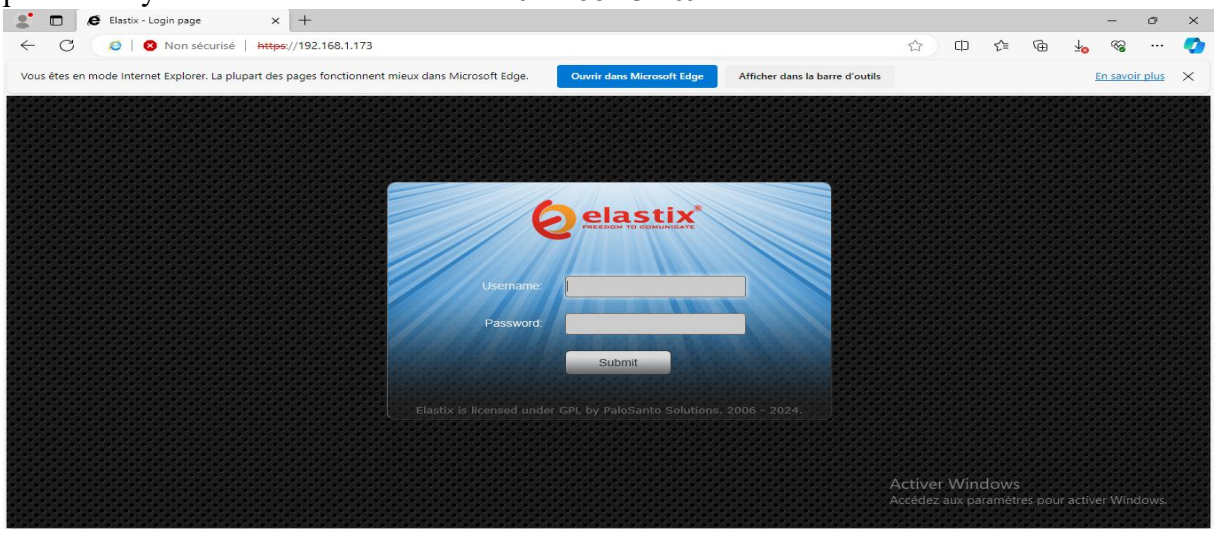
[root@Elastix ~]# bash
[root@Elastix ~]# yum install iptables
Loaded plugins: fastestmirror
Determining fastest mirrors
Could not retrieve mirrorlist http://mirrorlist.centos.org/?release=5&arch=i386&
repo=addons error was
[Errno 4] IOError: <urlopen error (-3, 'Temporary failure in name resolution')>
Error: Cannot find a valid baseurl for repo: addons
[root@Elastix ~]# bash service iptables start
```

figure 3.3.3 1 firewall installation

3.4 TELEPHONE SERVICE CONFIGURATION

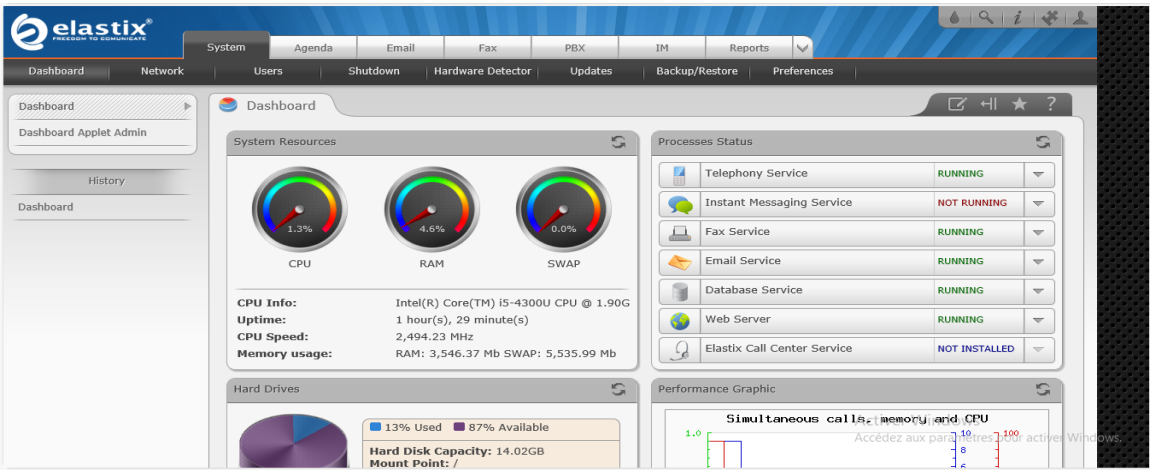
After the download of packages is complete, we will proceed to the steps of configuring the telephone service.

- We will enter ELASTIX configuration interface mode by entering the IP address provided by the server. In our case it's 192.168.43.169



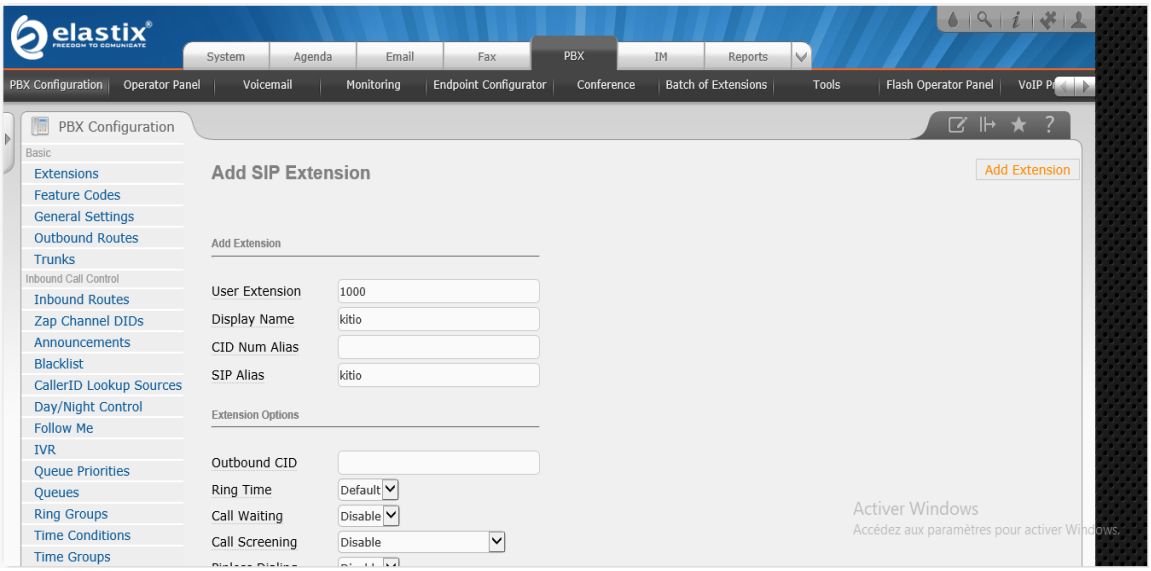
3.4 a 1: Elatix Authentication

- Then we will authenticate to access the graphical interface, it is in this interface that we can add user accounts by opening the file



3.4 a 2 :Graphical Interface of Elastix

- After adding user accounts, it will be necessary to assign a call number to each account and authorize each user to make a call



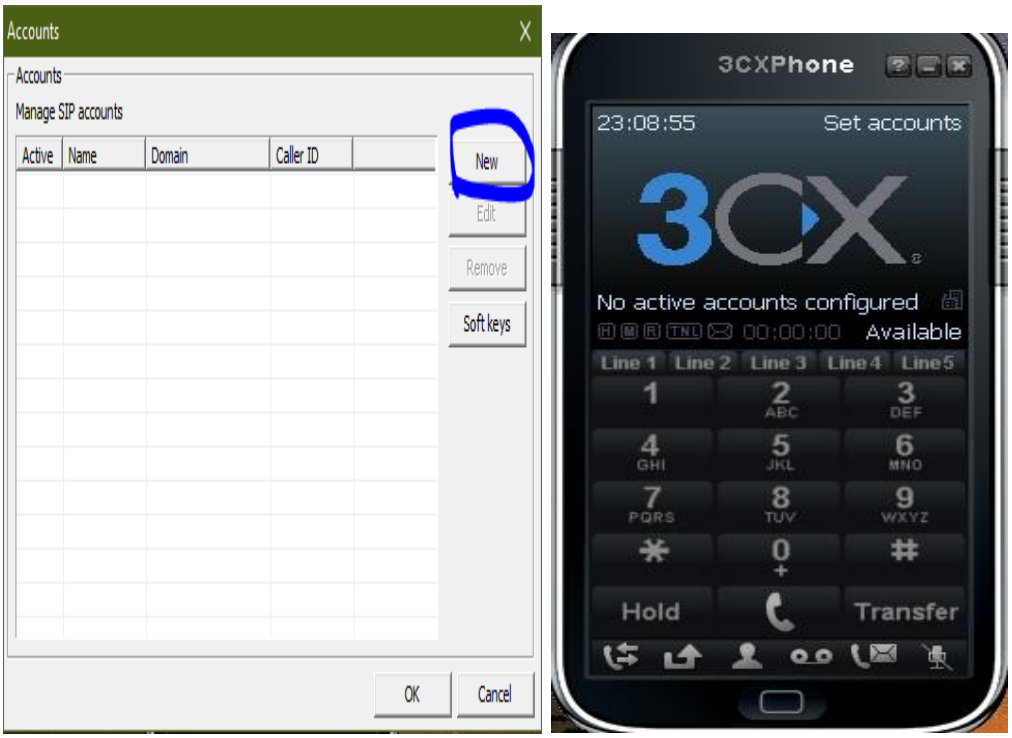
3.4 a 3: SIP Configuration

- After adding users, it is necessary to restarts Elastixs Server

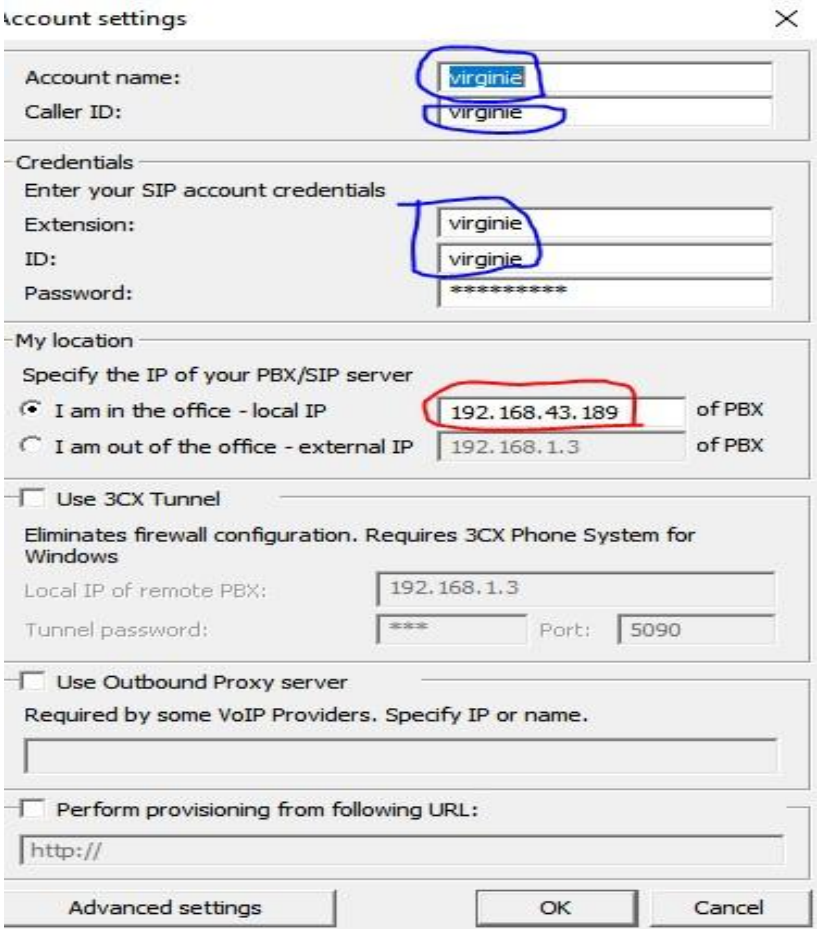
3.5 CONFIGURING CLIENTS

This involves the setting up of IP Phones, soft-phones, Wi-Fi IP phones, and any other terminal that can connect to the IP. In our case we will configure he soft phones. The

configuration of softphones is different from one and another but all obey the same principal. We are going to configure 3cxphones.



3.5 1 : 3cx phone configuration



3.5 2 : 3cx phone user settings

CONCLUSION

IP telephony has exponentially improved traditional or analog telephony. It uses the same transmission media as a standard computer network because in the IP network, the data path is not different because it uses the protocols of the OSI model layers. Some security aspects were also implemented such as: firewall.

CHAPTER 4: RESUSLTS, CONCLUSION AND RECOMMENDATIONS

4.1: Results and Interpretation:

The installation of the Elastix Server for the VoIP system has been completed, and the results are presented in this chapter. The successful deployment of the VoIP system is a critical component of the organization's communication infrastructure, and it is essential to evaluate its performance, functionality, and security. As a vital part of our organization's communication infrastructure, it's crucial to assess how well the VoIP system is working and identify areas where we can make improvements. These resulys can be mentioned below ;

Successful Installation :

- Elastix Server was successfully installed and configured for the VoIP system.
- The installation process was completed within the expected timeframe.

Functionality Testing :

The VoIP system was tested for functionality, including:

- Making and receiving calls
- Call routing and dial plans
- Voicemail and email notifications

All features were found to be working as expected.

Performance Evaluation :

- The VoIP system's performance was evaluated, including:
  Call quality and latency
  System uptime and reliability
- The system was found to be performing well, with minimal latency and no downtime.

Security Measures :

- Security measures were implemented, including:
  Firewall configuration
  Encryption and authentication protocols
- The system was found to be secure and compliant with industry standards.

User Feedback :

- Users were trained on the new VoIP system and provided feedback on its functionality and performance.
- Feedback was positive, with users reporting improved call quality and ease of use.

Final Steps :

- The VoIP system will be monitored and maintained regularly to ensure optimal performance and security.
- Future upgrades and expansions will be planned and implemented as needed.

General Conclusion :

This internship at ETS YARO CONSULTING provided valuable hands-on experience in installing a VoIP system. The project involved assessing the network infrastructure, conducting a site survey, selecting a VoIP provider, and configuring the VoIP system. Through this experience, I gained practical knowledge of VoIP technology, network configuration, and security measures. The successful deployment of the VoIP system demonstrated the importance of careful planning, testing, and maintenance. This internship not only enhanced my technical skills but also provided insight into the consulting industry and the role of technology in business operations. Overall, it was a rewarding experience that has prepared me for future career opportunities in the field of telecommunications and IT consulting.

As recommendation, based on the VoIP system we implemented , We recommend the company ETS Yaro consulting to emphasize on the following aspects, especially in the security enhancement . These aspects include:

Security Enhancements :

1. Implement regular software updates: Regularly update the VoIP system's software to ensure that any security vulnerabilities are patched.

2. Use strong passwords and authentication: Implement strong passwords and authentication protocols to prevent unauthorized access to the VoIP system.

3. Configure firewalls and intrusion detection systems: Configure firewalls and intrusion detection systems to monitor and block suspicious traffic.

Quality of Service (QoS) Improvements :

1. Prioritize VoIP traffic: Prioritize VoIP traffic to ensure that it receives sufficient bandwidth and quality of service.

2. Implement Quality of Service (QoS) policies: Implement QoS policies to ensure that VoIP traffic is treated with high priority.

Scalability and Redundancy

1. Implement redundancy: Implement redundancy in the VoIP system to ensure that it remains available in case of hardware or software failures.

2. Plan for scalability: Plan for scalability to ensure that the VoIP system can handle increased traffic and user growth.

User Training and Support :

1. Provide user training: Provide user training on the VoIP system's features and functionality.

2. Offer technical support: Offer technical support to users to ensure that any issues are resolved quickly and efficiently.

Monitoring and Maintenance :

1. Regularly monitor the VoIP system: Regularly monitor the VoIP system to ensure that it is performing optimally.

2. Perform regular maintenance: Perform regular maintenance tasks, such as backups and software updates, to ensure that the VoIP system remains stable and secure.

Also, we recommend the company to implement additional festures, such as ;

- **Implement advanced features**: Implement advanced features, such as call recording, call forwarding, and voicemail-to-email.
- **Integrate with other systems**: Integrate the VoIP system with other systems, such as CRM and ERP systems, to enhance functionality and productivity.